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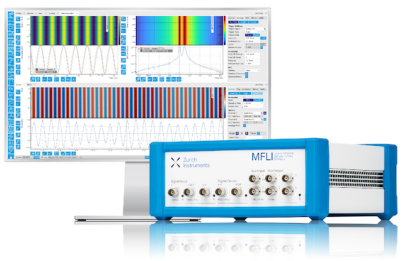
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Application of direct potential fitting to line position data for the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states of LiH

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A collection of 9089 spectroscopic LiH line positions, of widely varying precision, which sample 84.9% and 98.6% of the A and X state well depths, respectively, have been employed in a direct least-squares fit of the effective potential energy and Born-Oppenheimer breakdown functions for the two states. For the four isotopomers ^{6}LiH, ^{7}LiH, ^{6}LiD, and ^{7}LiD, the data comprise both pure rotational and vibration-rotational transitions within the ground state, as well as rotationally resolved transitions in the A-X system. Despite the unusual shape and associated anomalous properties of the A state potential, no special features or considerations were required in the direct potential fitting approach. The reduced standard deviation of the fit was close to unity, indicating that the quantum mechanical eigenvalues calculated from the fully analytical functions of the Hamiltonians of the two states, which are characterized by a total of only 53 fitted parameters, represent the line positions, on average, to within the estimated uncertainties. A quantum mechanical calculation of the molecular constants G_v, B_v, D_v, H_v, L_v, M_v, N_v, and O_v from the fitted potential for the A state of ^{7}LiH confirms that the usual polynomial expansion in $J(J+1)$ is an unsatisfactory representation for the rotational terms of the lowest vibrational levels. © *2004 American Institute of Physics.* [DOI: 10.1063/1.1788659]

I. INTRODUCTION

Over the last three decades, LiH has been the subject of many spectroscopic and theoretical investigations.[1] It is the lightest of stable neutral heteronuclear diatomic molecules, so that significant Born-Oppenheimer (BO) breakdown effects are expected. This has certainly been one of the motivations for the acquisition of extensive spectroscopic data for the four stable isotopomers, ^{6,7}Li^{1,2}H, for transitions within the $X\,^1\Sigma^+$ ground state, as well as for the $A\,^1\Sigma^+$-$X\,^1\Sigma^+$ system; with such data, it is possible to investigate the isotopic variation of BO breakdown over wide radial ranges in two electronic states of this very light molecule. The highly unusual shapes of the potential energy curves of the $A\,^1\Sigma^+$ states of the alkali metal hydrides in general, but especially that of the A state of LiH, have also long been a source of fascination to experimentalists and theoreticians alike. The potential of LiH(A) is exceptionally flat over a wide radial range around the equilibrium position R_e, and this highly anharmonic, nonparabolic nature leads to the well-known and unusual negative values of the anharmonicity constant $\omega_e x_e$ and the vibration-rotational coupling constant α_e. Mulliken[2] was first to provide a comprehensive and essentially correct explanation for such behavior. As shown in Fig. 1, Mulliken's description of the A and X states of LiH involves an avoided crossing at $R \approx 3$ Å between the strongly bound ionic state dissociating to $\mathrm{Li}^+(^1S)+\mathrm{H}^-(^1S)$ and the weakly bound covalent state dissociating to ground state atoms, $\mathrm{Li}(2\,^2S_{1/2})+\mathrm{H}(1\,^2S_{1/2})$. However, a second avoided crossing before the ionic limit ensures that the A state actually dissociates to neutral atoms, $\mathrm{Li}(2\,^2P_{1/2})+\mathrm{H}(1\,^2S_{1/2})$, rather than to ions. According to this picture, the X state is ionic in character at low energies around R_e, but there is a gradual change toward covalent character with increasing internuclear separation toward dissociation. The opposite pattern holds for the A state (except that as indicated above, the eventual dissociation products are neutral atoms). The most direct confirmation of this description is provided by the several determinations[3–5] of the large dipole moment ($p_0 = 5.882$D) in $v=0$ of the X state, by the relatively small and rapidly varying dipole moments[6,7] for low-v levels of the A state and by *ab initio* calculations[8] of the $p(R)$ functions for the two states. In addition, an explanation is afforded for the observed dissociation energy of LiH(X), which is anomalously small relative to that expected for dissociation to the $\mathrm{Li}^+ + \mathrm{H}^-$ ion pair limit.

One of the less familiar consequences of the unusual shape of the LiH(A) potential energy curve was identified by Orth and Stwalley[9] in their detailed rotational analysis of numerous bands of the A-X system of ^{7}LiH. It was found that the traditional representation of the rotational terms of the A state by a polynomial in $J(J+1)$,

$$F_v(J) = B_v J(J+1) - D_v[J(J+1)]^2 + H_v[J(J+1)]^3, \tag{1}$$

became increasingly unsatisfactory with *decreasing* v. In fitting the data to Eq. (1), it was necessary to define values of J_{max} such that data for $J > J_{max}$ were excluded. For $v=0$, 1, and 2 the values of J_{max} were 15, 22, and 29, respectively. Such an unusual result can be appreciated by considering effective potentials $U_J(R)$ for specific J values. Figure 2, for example, shows the $U_J(R)$ potentials for $J=0$ and $J=33$

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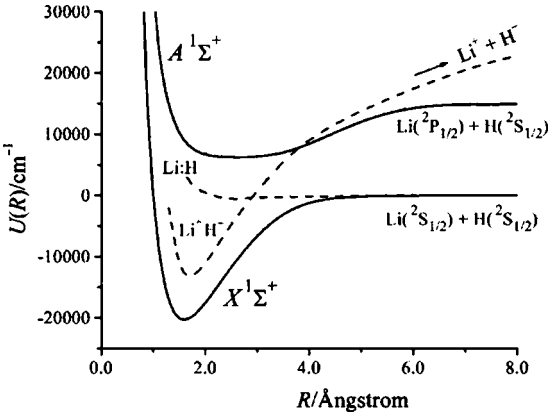


FIG. 1. Potential energy curves for the $A\ ^1\Sigma^+$ and $X\ ^1\Sigma^+$ states of LiH. The solid curves represent the potentials determined in the present work. Following Mulliken (Ref. 2), these are considered to arise from an avoided crossing between the zero-order diabatic Li^+H^- and Li:H potentials, shown as dashed curves. A second crossing near 5 Å (not shown) ensures that the *A* state actually dissociates to neutral atoms, rather than to Li^+ and H^-.

obtained in the present work, where $U_J(R)$ is calculated by adding the rotational energy to the rotationless $J=0$ potential $U_0(R)$,

$$U_J(R)=U_0(R)+B(R)J(J+1); \tag{2}$$

the rotational parameter is defined as $B(R)=\hbar^2/(2\mu R^2)$. The $U_{33}(R)$ potential is considered as $J=33$ is the highest J level observed in $\nu_A=0$ for ^{7}LiH. Figure 2 shows that there is a very large increase in R_e from 2.597 to 3.238 Å between the $U_0(R)$ and $U_{33}(R)$ potentials. For $\nu_A=0$, the inner turning point for $J=33$ is very close to the outer turning point for $J=0$. Given such a dramatic rotational distortion effect, it is not surprising that Eq. (1) fails for the lower vibrational levels, especially for $\nu_A=0$. For the same reason, up to 56 Y_{kl} coefficients were needed[10] for a single isotopomer fit when representing *A*-state vibration-rotational term values by a Dunham expansion,

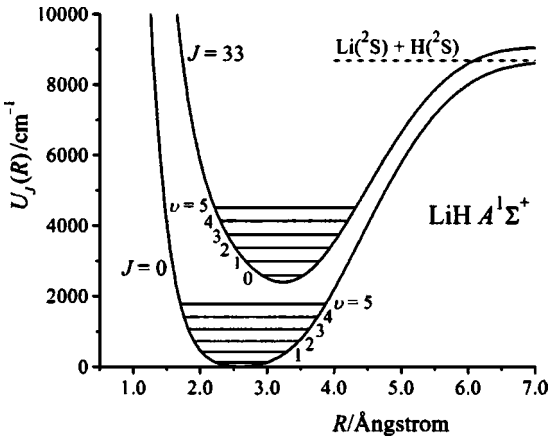


FIG. 2. The effect of rotational distortion in the $A\ ^1\Sigma^+$ state of LiH. For $J=33$ rotational excitation, the minimum of the effective potential is shifted by about 0.64 Å from the minimum of the rotationless potential. For $J=0$, the vibrational spacings $\Delta G_{\nu+1/2}$ are seen to *increase* (from 281 to 366 cm^{-1} for $\nu=0$–4), whereas the corresponding spacings in the $J=33$ potential decrease in a normal fashion from 389 to 379 cm^{-1}.

$$E_{\nu J}=\sum_{k=0}\sum_{l=0} Y_{kl}(\nu+\tfrac{1}{2})^k[J(J+1)]^l, \tag{3}$$

and for a less extensive and less accurate data set than that considered in the present work. In addition to these problems, any unified description of the data for all four isotopomers requires a consideration of BO-breakdown effects, as was first recognized by Vidal and Stwalley.[10] This work[10] included a pioneering effort to determine radially dependent "adiabatic" correction functions $U^k(R)$ for the $k=$Li and H centers and for both the *A* and *X* states. The authors employed an *inverted perturbation approach* (IPA), which was first proposed for diatomic systems by Kosman and Hinze[11] and is similar to the method employed earlier by Le Roy and Van Kranendonk[12] for determination of the H_2-Rg potential surface, where Rg represents a rare gas element. While the work of Vidal and Stwalley[10] convincingly demonstrated significant effects of the $U^k(R)$ correction functions, the limited computing power at the time necessarily required an approach quite inferior to that of a contemporary analysis. Apart from the limitations of the IPA procedure, in which unknown correction functions are represented by a truncated set of standard functions and are obtained in an unsatisfactory pointwise form, it was necessary to perform *separate* fits for the two states; this required the determination of intermediate sets of term values, whereas a more satisfactory procedure would employ directly the measured line positions. The term values for each state were represented by Dunham-like expansions, in which correction functions for BO breakdown were added to the standard form of Eq. (3). The standard deviation of the global fit was significantly larger than the typical precisions of the line positions. In summary, the derived BO-potential and adiabatic correction functions did not adequately represent the line position data. A few years later, the same methodology was employed[13] to obtain improved results by taking account of newly available fluorescence data; these data provided term values for the ground electronic state over a much wider range of vibrational levels. At about the same time, Hadinger and Tergiman[14] applied a novel algebraic transformation employing the earlier term value expansion coefficients of Vidal and Stwalley[10] so as to obtain the *X*-state BO potential and the adiabatic corrections in analytical form. In fact, the determination of $U_X^{Li}(R)$ by Hadinger and Tergiman[14] is the only estimate of this function prior to the present work. For the *X*-state, Stwalley and co-workers[10,13] were able to determine only $U_X^H(R)$, which describes the much larger BO-breakdown effects for H/D isotopic substitution.

Over the last decade or so, considerable progress has been made in the development of *direct potential fitting* (DPF) methods for the analysis of accurate and extensive data sets of diatomic molecules. Line positions are employed directly, and there is an emphasis on representation of these data by the derived radial Hamiltonians to within the associated uncertainties. An overview of various algebraic and numerical methods has been provided by Coxon and Molski.[15] In particular, it was found that for numerous molecules, results published by Ogilvie from an algebraic approach[16] known as *Radiatom* are unreliable and discrepant with other

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treatments. The nature of the problem with the computer program of Ref. 16 was identified recently by Le Roy.[17] The numerical method employed in the present DPF approach was developed in collaboration with Hajigeorgiou,[18–20] and permits the inclusion of data from electronic transitions, from which the radial functions of two electronic states are determined simultaneously. Recent applications of this type include the A-X system of BeH^+ (Ref. 21) and the B-X system of HCl.[22] Analyses of vibration-rotational and pure rotational data within LiH(X) have been published previously.[15,23]

The purpose of the present work is to apply our modern numerical DPF approach to the available line position data for the A and X states of LiH. Since the 1986 IPA analysis of Chan *et al.*,[13] considerable amounts of new data have become available,[24–29] most of which either sample previously unobserved levels or are much improved in quality. An additional motivation for the present work was to provide a better description of the significant differences between the $U^{\mathrm{H}}_{X,A}(R)$ correction functions for the A and X states of the IPA analysis[13] and the *ab initio* calculations of the adiabatic correction functions of Bishop and Cheung.[30,31]

II. THEORY AND MODELING

The numerical approach has been described in detail by Coxon and Hajigeorgiou,[22,32] where for HCl and CO separate least-squares fits were performed with the radial operator obtained[33,34] from Watson's work[35] and with that described by Herman and Ogilvie.[36] In the absence of any information on the rotational g_J factor for the A state of LiH, only the first approach was adopted in the present work. The effective Hamiltonian[33,34] for a diatomic isotopomer i in a $^1\Sigma$ electronic state is given by

$$\hat{H}_i = (2\mu_i^{\mathrm{at}})^{-1} P_R^2 + U_i^{\mathrm{eff}}(R) + B_i(R)[1+q_i(R)] \times J(J+1), \tag{4}$$

where μ_i^{at} is the reduced mass defined in terms of atomic masses. P_R^2 represents the nuclear kinetic energy operator, $U_i^{\mathrm{eff}}(R)$ is the effective potential energy, $q_i(R)$ describes BO-breakdown effects, and $B_i(R)=\hbar^2/(2\mu_i^{\mathrm{at}}R^2)$ is the radial part of the rotational energy operator.

The A- and X-state effective potential energy functions for an isotopomer of LiH can be partitioned as follows:

$$U_i^{\mathrm{eff}}(R) = U^{\mathrm{BO}}(R) + \frac{m_e U^{\mathrm{Li}}(R)}{M_i^{\mathrm{Li}}} + \frac{m_e U^{\mathrm{H}}(R)}{M_i^{\mathrm{H}}}, \tag{5}$$

where m_e is the electron rest mass, $U^{\mathrm{BO}}(R)$ is the BO-potential energy function, and $U^{\mathrm{Li}}(R)$ and $U^{\mathrm{H}}(R)$ are isotopically independent BO-breakdown functions that contain adiabatic (diagonal) and J-independent nonadiabatic (off-diagonal homogeneous $^1\Sigma\sim{}^1\Sigma$) corrections. In the absence of constraints given by the electric dipole moment $p(R)$ and rotational $g_J(R)$ functions, these diagonal and off-diagonal effects are not uniquely separable using field-free data alone.[17,37]

The J-dependent breakdown corrections resulting from homogeneous and heterogeneous interactions, again combined in a generally indeterminable proportion, are described by two atom-centered mass-independent functions $q^{\mathrm{Li}}(R)$ and $q^{\mathrm{H}}(R)$, which contribute to the isotopically dependent function $q_i(R)$,

$$q_i(R) = \frac{m_e q^{\mathrm{Li}}(R)}{M_i^{\mathrm{Li}}} + \frac{m_e q^{\mathrm{H}}(R)}{M_i^{\mathrm{H}}}. \tag{6}$$

The radial Hamiltonian operator of Eq. (4), together with the auxiliary functions of Eqs. (5) and (6), requires spectroscopic data that sample isotopic substitution at both atomic centers in order to achieve reliable determinations of the various isotopically invariant functions. The mathematical models employed in representing the various radial functions are of critical importance to the success of the overall procedure.

The model used to represent the rotationless potential energy is the MLJ oscillator of Hajigeorgiou and Le Roy,[38] which is also known as the Morse/Lennard-Jones oscillator,[39]

$$U(R) = D_{\mathrm{e}}\left[1-\left(\frac{R_{\mathrm{e}}}{R}\right)^n e^{-\phi(R)z}\right]^2, \tag{7}$$

where n is the inverse power of the leading term in the inverse power series for the potential energy in the long-range region,

$$U(R) = D_{\mathrm{e}} - \frac{C_n}{R^n} - \cdots, \tag{8}$$

$z=2(R-R_{\mathrm{e}})/(R+R_{\mathrm{e}})$ is a reduced variable for the internuclear separation (R), and $n=6$ for the X and A states of LiH.

The MLJ-oscillator model is best employed when the leading coefficient C_n in Eq. (8) is already known, permitting its use as a valuable long-range constraint. This is possible because of the transformation of the potential energy in the long region to a functional form equivalent to Eq. (8). Thus, the asymptotic value of $\phi(R)$ as $R\to\infty$ can be constrained to the value given by

$$\phi(\infty) = \frac{1}{2}\ln\left(\frac{2D_{\mathrm{e}}R_{\mathrm{e}}^n}{C_n}\right). \tag{9}$$

In order that the asymptotic limit $\phi(\infty)$ is approached smoothly and sensibly, $\phi(R)$ is defined[22,32] by a polynomial expansion in z, in combination with a switching function $f_{\mathrm{sw}}(R)$, such that

$$\phi(R) = f_{\mathrm{sw}}(R)\sum_{m=0}\phi_m z^m + [1-f_{\mathrm{sw}}(R)]\phi(\infty), \tag{10}$$

with

$$f_{\mathrm{sw}}(R) = [1+e^{\delta(R-R_{1/2})}]^{-1}. \tag{11}$$

δ controls the damping strength and $R_{1/2}$ is the R value at the 50% switching point. In the present work, the $R_{1/2}$ parameters are constrained at values corresponding to the largest R values sampled by the data, as indicated by the wave functions of the highest-lying (v,J) levels; the values of δ are fixed at values based on the experience from previous work, though in fact the fits are not sensitive to any reasonable choice. Thus while individual parameters ϕ_m may take on slightly modified values to accommodate different choices

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for δ (and $R_{1/2}$), the radial variations of the fitted functions (including the contribution from switching functions) are insensitive, by and large, to the exact values selected for the switching function parameters.

The potential correction radial functions $U^k(R)$ in Eq. (5) and $q^k(R)$ in Eq. (6) are also controlled at large R by separate switching functions, such that

$$U^k(R) = f_{\mathrm{sw}}^{U^k}(R) \sum_{m=1} u_m^k z^m + u_\infty^k \left[1 - \frac{f_{\mathrm{sw}}^{U^k}(R)}{f_{\mathrm{sw}}^{U^k}(R_\mathrm{e})} \right], \tag{12}$$

and

$$q^k(R) = f_{\mathrm{sw}}^{q^k}(R) \sum_{m=1} q_m^k x^m, \tag{13}$$

where $k = \mathrm{Li}$ or H and x is the reduced variable $(R - R_\mathrm{e})/R_\mathrm{e}$. Equation (12) is consistent with the arbitrary condition $U^k(R_\mathrm{e}) = 0$, and Eq. (13) ensures that the $q^k(R)$ functions tend asymptotically to zero as $R \to \infty$, as is expected theoretically.[40]

III. LEAST-SQUARES FITTING

In the previous analyses of the data sets of HeH^+ (Ref. 20) and HCl,[22] both involving four isotopomers, the usual application of Eq. (5) led to a representation in which the BO-potential-energy function was the root potential, from which effective potentials for individual isotopomers could be constructed. In the case of CO, Coxon and Hajigeorgiou[32] also implemented the approach proposed by Le Roy[41] for situations where the data set of one isotopomer, known as the *dominant* isotopomer, outweighs the data sets of other isotopomers in the extent of vibration-rotational excitation and/or in the quality of the measurements. For LiH, the dominant isotopomer is clearly identified as $^7\mathrm{Li}^1\mathrm{H}$. There is an important difference, however, between our convention and that of Le Roy[41] in the definitions of the parameter sets employed for the effective internuclear potentials $U^{\mathrm{eff}}(R)$ and the BO-breakdown correction function $q(R)$. With the functions for isotopomer i of LiH defined by

$$U_i^{\mathrm{eff}}(R) = U_1^{\mathrm{eff}}(R) + m_e \frac{(M_1^{\mathrm{Li}} - M_i^{\mathrm{Li}})}{M_1^{\mathrm{Li}} M_i^{\mathrm{Li}}} U^{\mathrm{Li}}(R) + m_e \frac{(M_1^{\mathrm{H}} - M_i^{\mathrm{H}})}{M_1^{\mathrm{H}} M_i^{\mathrm{H}}} U^{\mathrm{H}}(R) \tag{14}$$

and

$$q_i(R) = \frac{m_e}{M_i^{\mathrm{Li}}} q^{\mathrm{Li}}(R) + \frac{m_e}{M_i^{\mathrm{H}}} q^{\mathrm{H}}(R), \tag{15}$$

where the subscript "1" refers to the dominant isotopomer, an exact equivalence is preserved between the radial functions $U^{\mathrm{Li}}(R)$, $U^{\mathrm{H}}(R)$, $q^{\mathrm{Li}}(R)$, and $q^{\mathrm{H}}(R)$ of Eqs. (14) and (15) and those of Eqs. (5) and (6). They differ from Le Roy's functions,[41] which are dependent on the choice of reference isotopomer (1), by simple mass scaling factors.

Full details of the numerical procedure have been described previously.[22,23,32] Briefly, we employ a weighted non-linear least-squares fitting procedure using the residuals between calculated and observed line positions and employing $(a_k)^{-2}$ as weights, where a_k are the estimated uncertainties of the measured line positions. The partial derivatives required for iterative least-squares adjustment of parameters p_k are obtained as Hellmann-Feynman integrals,

$$\frac{\partial E_{vJ}}{\partial p_k} = \langle \psi_{vJ} | \frac{\partial \hat{H}^{\mathrm{eff}}(R)}{\partial p_k} | \psi_{vJ} \rangle. \tag{16}$$

IV. SPECTROSCOPIC DATA FOR THE $X\,^1\Sigma^+$ AND $A\,^1\Sigma^+$ STATES OF LiH

A. Line positions

The data set employed in the present work is summarized in Table I and contains 9089 measured line positions (3161, 2331, 1747, and 1850 lines for ^{7}LiH, ^{6}LiH, ^{7}LiD, and ^{6}LiD, respectively). This is the most extensive set of data for LiH analyzed to date; a wide range of energy levels is sampled in the A and X states, ranging over 84.9% and 98.6% of the well depths, respectively.

Precise pure rotational data for the low vibrational levels of all four isotopomers are available from the work of Pearson and Gordy,[42] Plummer, Herbst, and Lucia,[43] Matsushima *et al.*,[26] and Bellini *et al.*[27] More extensive, but less accurate, pure rotational data have been obtained by Maki, Olson, and Thompson[25] and recently by Dulick *et al.*[28] Vibration-rotational data for low vibrational levels of the X state have been obtained using Fourier transform infrared (FTIR) absorption by Maki *et al.*[25] and by Dulick *et al.*[28] with measurement uncertainties of 0.000 5–0.001 5 cm^{-1}. Slightly less precise data (0.002 cm^{-1}) were obtained by Yamada and Hirota[24] using IR diode laser absorption. None of these IR data sets were available in the IPA analysis of Chan *et al.*[13]

Most of the data on the A-X system of LiH have been obtained in both absorption and emission using conventional spectrographic techniques. The very early, but excellent, data of Crawford and Jorgensen[44] were extended in the detailed studies of Stwalley and co-workers.[9,45,46] Recently, however, more accurate, but less extensive, data were obtained by Bouloufa *et al.*[29] using a sub-Doppler cw-laser absorption technique. These data sets are complemented by much less accurate measurements from laser fluorescence studies.[45,47,48] These data are valuable, however, as they provide the only information for the highest v levels of the ground state. It should be mentioned that the data of Chan[48] for the highest level of ^{7}LiH(X), $v=22$, were not employed in the present analysis. We find that the $J=10$ state in $v=22$ is quasibound with a level width of about 1.4 cm^{-1}, so that it would seem unlikely that the $P(10)$ line of the $A \to X$ 10-22 band could be measured reliably. Although the corresponding $R(8)$ line does not involve a quasibound level, we find that this transition appears unreliable also. While the $R(8)$ line can be included satisfactorily in a least-squares DPF analysis of the $^7\mathrm{Li}^1\mathrm{H}$ data alone, large residuals were obtained for transitions involving the highest v'' levels of all isotopomers when this line was included in the global, multiisotopomer fit.

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TABLE I. Experimental line position data for the $A\,^1\Sigma^+$ and $X\,^1\Sigma^+$ states of ^{6,7}LiH and ^{6,7}LiD. For each data set, the table shows the number of line positions employed in the least-squares fits of the present work; $\Delta\nu$ is the change in vibrational quantum number for transitions within the X state; the maximum vibrational and rotational quantum numbers for the set are given as $\nu_{\max}$ and $J_{\max}$.

Isotopomer	Type	Uncertainty $(\mathrm{cm})^{-1}$	$\Delta\nu$	ν_A	ν_X	$J_{\max}$	No. of lines	Authors	Reference	Year
^{6}LiH(X)	Submillimeter	1.67×10^{-6}	0		0	0	1	Plummer, Herbst, and De Lucia	43	1984
^{6}LiH(X)	Far-IR absorption	$(4.3-10.0)\times10^{-6}$	0		0-1	5	4	Bellini *et al.*	27	1995
^{6}LiH(X)	Far-IR absorption	4.5×10^{-6}	0		0	9	8	Matsushima *et al.*	26	1994
^{6}LiH(X)	FT/IR emission	0.000 5	0		0-1	26	36	Dulick *et al.*	28	1998
^{6}LiH(X)	FT/IR absorption	0.000 6	0		0-1	19	20	Maki, Olson, and Thompson	25	1990
^{6}LiH(X)	FT/IR emission	0.000 5–0.001	1		0-3	20	71	Dulick *et al.*	28	1998
^{6}LiH(X)	FT/IR absorption	0.000 8	1		0-3	20	66	Maki, Olson, and Thompson	25	1990
^{6}LiH(X)	IR diode laser absorption	0.002	1		0-1	1	1	Yamada and Hirota	24	1988
^{6}LiH(A-X)	Sub-Doppler absorption	0.005–0.01		1-10	0-2		104	Bouloufa *et al.*	29	2000
^{6}LiH(A-X)	Visible emission	0.10		1-16	0-7		2001	Li	45	1976
^{6}LiH(A-X)	Laser fluorescence	0.10–0.30		12	6-21		19	Chan	48	1985
^{7}LiH(X)	Submillimeter	1.67×10^{-6}	0		0-1	0	2	Plummer, Herbst, and De Lucia	43	1984
^{7}LiH(X)	Far-IR absorption	$(1.5-6.0)\times10^{-6}$	0		0-2	6	8	Bellini *et al.*	27	1995
^{7}LiH(X)	Far-IR absorption	4.5×10^{-6}	0		0-1	10	17	Matsushima *et al.*	26	1994
^{7}LiH(X)	FT/IR emission	0.000 5	0		0-3	29	77	Dulick *et al.*	28	1998
^{7}LiH(X)	FT/IR absorption	0.000 6	0		0-2	21	33	Maki, Olson, and Thompson	25	1990
^{7}LiH(X)	FT/IR emission	0.000 5–0.001 5	1		0-5	28	176	Dulick *et al.*,	28	1998
^{7}LiH(X)	FT/IR absorption	0.001	1		0-3	27	87	Maki, Olson, and Thompson	25	1990
^{7}LiH(X)	IR diode laser absorption	0.002	1		0-2	11	14	Yamada and Hirota	24	1988
^{7}LiH(X)	FT/IR emission	0.001–0.002	2		0-6	25	118	Dulick *et al.*	28	1998
^{7}LiH(A-X)	Sub-Doppler absorption	0.005		1-10	0-3		260	Bouloufa *et al.*	29	2000
^{7}LiH(A-X)	uv/vis absorption	0.05		0-15	0-2		641	Crawford and Jorgensen	44(a)	1935
^{7}LiH(A-X)	uv/vis absorption	0.05		1-4	2-3		261	Crawford and Jorgensen Orth and Stwalley	44(a) 9	1935 1979
^{7}LiH(A-X)	Visible emission	0.07		0-3	4-5		365	Li and Stwalley Orth and Stwalley	46 9	1978 1979
^{7}LiH(A-X)	Visible emission	0.07		0-7	1-12		965	Orth and Stwalley	9	1979
^{7}LiH(A-X)	Laser fluorescence	0.15		16	8-20		15	Chan	48	1985
^{7}LiH(A-X)	uv absorption	0.30		16-20	0		122	Rafi, Iqbal, and Baig	49	1983
^{6}LiD(X)	mm-wave abs.	1.67×10^{-6}	0		0-1	0	2	Pearson and Gordy	42	1969
^{6}LiD(X)	Submillimeter	1.67×10^{-6}	0		0	1	1	Plummer, Herbst, and De Lucia	43	1984
^{6}LiD(X)	Far-IR absorption	$(4.3-6.3)\times10^{-6}$	0		0	10	2	Bellini *et al.*	27	1995
^{6}LiD(X)	FT/IR emission	0.000 5	0		0-1	36	38	Dulick *et al.*	28	1998
^{6}LiD(X)	FT/IR absorption	0.000 6	0		0-2	28	48	Maki, Olson, and Thompson	25	1990
^{6}LiD(X)	FT/IR absorption	0.000 8–0.001	1		0-3	25	101	Maki, Olson, and Thompson	25	1990
^{6}LiD(X)	FT/IR emission	0.001	1		0-3	25	69	Dulick *et al.*	28	1998
^{6}LiD(X)	IR diode laser absorption	0.002	1		0-2	9	4	Yamada and Hirota	24	1988
^{6}LiD(A-X)	Visible emission	0.11–0.14		1-17	0-7		1585	Li	45	1976
^{7}LiD(X)	mm-wave absorption	1.67×10^{-6}	0		0-1	0	2	Pearson and Gordy	42	1969
^{7}LiD(X)	Submillimeter	1.67×10^{-6}	0		0-1	1	2	Plummer, Herbst, and De Lucia	43	1984
^{7}LiD(X)	Far-IR absorption	$(2.1-10.0)\times10^{-6}$	0		0-2	10	6	Bellini *et al.*	27	1995
^{7}LiD(X)	FT/IR emission	0.000 5	0		0-3	44	100	Dulick *et al.*	28	1998
^{7}LiD(X)	FT/IR absorption	0.000 6	0		0-2	30	42	Maki, Olson, and Thompson	25	1990
^{7}LiD(X)	FT/IR emission	0.000 5–0.001 5	1		0-6	26	198	Dulick *et al.*	28	1998
^{7}LiD(X)	FT/IR absorption	0.000 5–0.001	1		0-4	26	124	Maki, Olson, and Thompson	25	1990
^{7}LiD(X)	IR diode laser absorption	0.001 5	1		0-3	17	23	Yamada and Hirota	24	1988
^{7}LiD(A-X)	uv/vis absorption	0.03		1-18	0-3		1149	Crawford and Jorgensen	44(b)	1935
^{7}LiD(A-X)	Laser fluorescence	0.25		19	8-27		20	Chan	48	1985
^{7}LiD(A-X)	Laser fluorescence	0.3–1.5		10	0-26		41	Ennen, Fieldler, and Ottinger	47	1981
^{7}LiD(A-X)	Laser fluorescence	0.3–1.5		14	0-26		40	Ennen, Fieldler, and Ottinger	47	1981
	Total no. of lines						9089			

Data for high vibrational levels in the A state of ^{7}LiH have been reported by Rafi and co-workers.[49,50] The $\nu'\leftarrow\nu''=0$ absorption progression was extended[49] to $\nu'=20$, and ten years later, data for the $\nu'\leftarrow\nu''=1$ progression were obtained[50] up to $\nu'=26$. In the present work, we have been able to incorporate only the measurements from the $\nu''=0$ progression.[49] Even for the $\nu''=0$ data, we found it necessary to assign an uncertainty of 0.3 cm^{-1} that is considerably larger than the stated measurement precision[49] of ≈0.1 cm^{-1}. From the results of fits with the $\nu''=0$ data included, it was straightforward to calculate line positions for the $\nu''=1$ progression with $\nu'=16-20$. Comparison with the measurements of Ref. 50 revealed large differences, often of several cm^{-1} and with an average of about 1 cm^{-1}, much greater than the stated measurement precision of ≈0.05 cm^{-1}. In addition to this demonstrated unreliability of the $\nu''=1$ bands

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TABLE II. Fitted parameters for the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states of LiH. The effective potential energy curves of the X and A states of ^{7}Li^{1}H are defined via Eq. (7) with $n=6$ and the parameters δ, $R_{1/2}$, $\phi(\infty)$, ϕ_i, R_e, and D_e in columns 5–7. Parameters constrained at fixed values are given in square brackets. Values in parentheses are one standard error, in units of the least significant digit shown in regular type. Additional digits are given as subscripts in order that functions can be reconstructed to reproduce the experimental line positions.

		$X\,^1\Sigma^+$	$A\,^1\Sigma^+$		$X\,^1\Sigma^+$	$A\,^1\Sigma^+$
	δ	[2.5]	[2.5]	δ	[2.25]	[2.5]
	$R_{1/2}$ (Å)	[5.0]	[6.0]	$R_{1/2}$ Å	[5.15]	[6.0]
U^{Li}	u_∞^{Li}	[0.0]	[0.0]	$\phi(\infty)$	[0.367 22]	[1.005 6]
	$10^{-4}u_1$ cm^{-1}	$-0.77_{132}(1)$	$-7.6_{410}(3)$	ϕ_0	$-4.201\,467\,2_{169}(8)$	$-5.247\,1_{107}(1)$
	$10^{-4}u_2$ cm^{-1}	$6.13_{522}(5)$	$6.6_{401}(4)$	ϕ_1	$0.806\,68_{167}(2)$	$0.420\,0_{608}(8)$
				ϕ_2	$0.110\,48_{407}(8)$	$2.609_{314}(3)$
	δ	[2.5]	[3.0]	ϕ_3	$0.532\,5_{794}(7)$	$2.40_{383}(2)$
	$R_{1/2}$ Å	[5.25]	[5.3]	ϕ_4	$0.379_{195}(4)$	$-2.24_{705}(5)$
	$m_e u_\infty^{\mathrm{H}}$ (u cm^{-1})	[−11.6]	[−6.0]	ϕ_5	$0.25_{342}(2)$	$2.8_{523}(2)$
	$10^{-5}u_1$ cm^{-1}	$-1.022\,9_{927}(1)$	$0.097_{920}(9)$	ϕ_6	$0.24_{914}(7)$	$7.6_{719}(4)$
U^{H}	$10^{-5}u_2$ cm^{-1}	$1.513\,1_{552}(5)$	$0.230_{810}(8)$	ϕ_7	$2.0_{402}(1)$	$-34.5_{363}(11)$
	$10^{-5}u_3$ cm^{-1}	$-1.47_{383}(2)$	$-2.78_{135}(7)$	ϕ_8	$-1.0_{855}(6)$	$-46.7_{876}(16)$
	$10^{-5}u_4$ cm^{-1}	$1.39_{309}(6)$	$1.19_{045}(4)$	ϕ_9	$-9.2_{553}(11)$	$98.9_{092}(36)$
	$10^{-5}u_5$ cm^{-1}	$-0.67_{490}(5)$	$2.3_{807}(2)$	ϕ_{10}	$14.2_{154}(13)$	$118.4_{652}(34)$
				ϕ_{11}	$12.6_{523}(40)$	$-133.4_{379}(64)$
	δ	[2.5]		ϕ_{12}	$-34.8_{674}(39)$	$-119.9_{985}(28)$
	$R_{1/2}$ Å	[5.5]		ϕ_{13}	$15.7_{635}(13)$	$52.2_{058}(46)$
	q_∞^{Li}	[0.0]				
q^{Li}	q_1	$0.80_{118}(5)$		R_e Å	$1.595\,594\,16_{124}(5)$	$2.597\,04_{938}(5)$
	q_2	$-0.5_{774}(3)$		D_e cm^{-1}	[20 286.0]	[8 680.7]
	δ	[2.5]	[3.0]			
	$R_{1/2}$ Å	[5.5]	[6.5]	ΔT^{el} cm^{-1}		$26\,509.82_{381}(1)$
	q_∞^{Li}	[0.0]	[0.0]	Δd^{Li} (u cm^{-1})		$3.06_{339}(9)$
q^{H}	q_1	$0.723_{131}(9)$	$0.85_{712}(4)$	Δd^{H} (u cm^{-1})		$-5.67_{393}(2)$
	q_2	$-1.06_{353}(8)$				
	q_3	$1.51_{795}(9)$				

with $\nu'=16$–20, there are obviously incorrect assignments in the 22-1 and 23-1 bands, and as noted by the authors,[50] a curious absence of any lines with $J<12$ in the 25-1 and 26-1 bands. In summary, the reliability of the data of Ref. 50 is clearly very doubtful.

In the initial stages of the present work, a systematic discrepancy was immediately apparent between the line positions for ^{6}LiH of Li (Ref. 45) and the more accurate measurements obtained recently by Bouloufa *et al.*[29] On average, this discrepancy was 0.23 cm^{-1}. Accordingly, we have adjusted all the measurements of Li (Ref. 45) for ^{6}LiH by this amount. As the work progressed it became apparent that most of the residuals for ^{6}LiD between the observed and fitted line positions were systematically large and with the same sign. While there are no other data to compare with in the case of ^{6}LiD, it was concluded that the same 0.23 cm^{-1} offset should also be applied to most of the data of Li[45] for ^{6}LiD. With this change, a relatively small uncertainty of 0.11 cm^{-1} could be assigned to most of the ^{6}LiD data, similar to the 0.10 cm^{-1} error established for the ^{6}LiH data.[45]

B. Dissociation energies of the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states of LiH

From measurements of the convergence limit in the $B\,^1\Pi \leftarrow X\,^1\Sigma^+$ absorption bands of ^{7}LiH (Ref. 51) and ^{7}LiD,[52] Vidal and Stwalley[10] calculated the dissociation energies of these two isotopomers as $D_e(X)=20\,286.8$ and $20\,292.6$ cm^{-1}, respectively, and $D_e(A)=8680.7$ and 8683.7 cm^{-1}, respectively. It was concluded[10] that to within 0.1 cm^{-1}, the D_e values for both states of ^{6}LiH were the same as those of ^{7}LiH, with a similar result for ^{6}LiD and ^{7}LiD. Using values of T_e found in the present work from preliminary fits for individual isotopomers, we calculate the same D_e values as those of Ref. 10. The 5.8 cm^{-1} increase in $D_e(X)$ when D is substituted for H can be compared with the value of 4.65 cm^{-1} that can be obtained from the *ab initio* calculations of Bardo *et al.*[53] The present work employs the 5.8 and 3.0 cm^{-1} increases in $D_e(X)$ and $D_e(A)$ for H→D isotopic substitution as constraints in the $U^{\mathrm{H}}(R)$ correction functions for the X and A states. Similary, the $U^{\mathrm{Li}}(R)$ functions are constrained such that there is no change in $D_e(X)$ or $D_e(A)$ for ^{6}Li→^{7}Li substitution.

C. Long-range C_6 parameters for the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states of LiH

As indicated earlier, Eq. (9) permits the limiting value of $\phi(R)$ as $R\to\infty$ to be constrained if the C_n coefficient of the leading inverse power interaction expansion at long range is known. In the case of the LiH, $n=6$ for the A and X states. For the X state, consistent theoretical estimates have been determined as $C_6=66.65$,[54] 67.4(0.7),[55] and 66.5(5.3) (Ref. 56) a.u. The error in the result obtained by Stacey and

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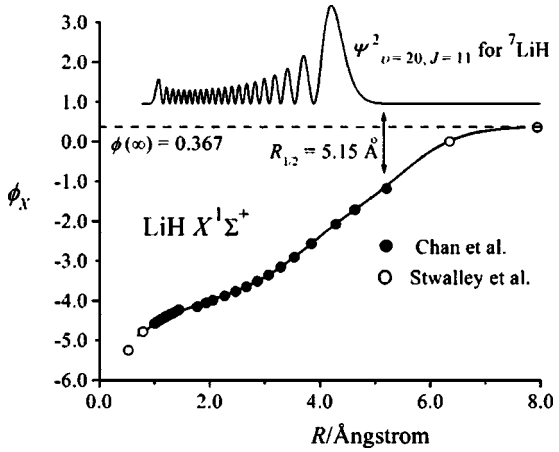


FIG. 3. The function $\phi_X(R)$ for the $X\,^1\Sigma^+$ state of ^{7}LiH. The curve is defined by the parameters in Table II for the potential determined in the present work with a switching function having $R_{1/2}=5.15$ Å. The points are from the IPA analysis of Chan *et al.* (Ref. 13) and the theoretical work of Stwalley *et al.* (Ref. 57). Also shown is ψ^2 for the $\nu_X=20$, $J=11$ state of ^{7}LiH, which is the highest-lying level for this isotopomer sampled by the data.

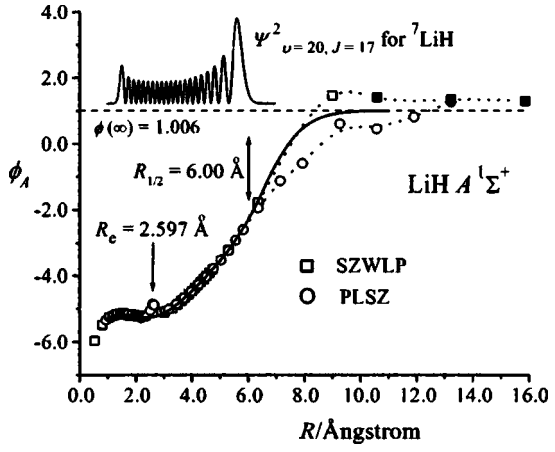


FIG. 4. The function $\phi_A(R)$ for the $A\,^1\Sigma^+$ state of ^{7}LiH. The curve is defined by the parameters in Table II for the potential determined in the present work with a switching function having $R_{1/2}=6.0$ Å. PLSZ refers to the *ab initio* calculations of Partridge *et al.* (Ref. 61) and SZWLP refers to the work of Stwalley *et al.* (Ref. 57). The three filled squares were calculated herein using the C_6 and C_8 parameters of Ref. 57. Also shown is ψ^2 for the $\nu_A=20$, $J=17$ state of ^{7}LiH, which is the highest-lying level sampled by the data.

Dalgarno[54] is stated as less than 2%. Using a method similar to that of Ref. 56, Stwalley *et al.*[57] calculated $C_6=147.9$ a.u. for the A state; we are not aware of a more recent calculation. The C_6 values of Refs. 54 and 57 were adopted in the present work.

V. RESULTS AND DISCUSSION OF THE GLOBAL LEAST-SQUARES FIT

Fundamental constants and atomic masses were taken from the most recent compilations.[58,59] Radial functions for the X state were obtained in the range 0.7–7.7 Å, with a mesh size of 0.001 75 Å; for the A state, the range was 1.0–7.5 Å with a mesh size of 0.001 625 Å; in both cases, there are 4001 points on the radial grid and tests were performed to ensure that the mesh sizes and radial ranges were defined such that error in the derived eigenvalues was negligible compared with the measurement uncertainties of the line positions. The 53 fitted parameters obtained from the global fit of the data in Table I for the four isotopomers are listed in Table II, along with 25 constrained parameters. A table of the observed and calculated line positions is available electronically as supplementary material.[60] The normalized standard deviation of the weighted least squares fit was $\hat{\sigma}=1.009$, very close to unity, as expected for a data set free of systematic error and with reliably estimated uncertainties.

From Eqs. (7)–(11), the $\phi(R)$ functions given by the fitted parameters in Table II define the effective potentials for the A and X states of the dominant isotopomer, ^{7}LiH. The effective potentials for other isotopomers can be found using Eq. (14) with the fitted $U^{\mathrm{Li}}(R)$ and $U^{\mathrm{H}}(R)$ functions. The $\phi(R)$ functions for the two states are shown in Figs. 3 and 4, together with the square of the radial wave functions for the highest-lying vibration-rotational states. Figure 3 demonstrates the approximately linear variation of $\phi_X(R)$ over most of the radial range of the outer limb of the potential; also, it is apparent that the effect of the switching function with $R_{1/2}=5.15$ Å is to impose an asymptotic approach to the long-range limit, $\phi_X(\infty)\approx 0.367$. Figure 3 also shows the excellent agreement with the IPA potential of Chan *et al.*[13] and with the two points at the largest R values calculated by Stwalley *et al.*[57] using only theoretical long-range coefficients C_6, C_8, and C_{10}. It should be noted that the curve in Fig. 3 obtained in the present work for the small R range between 0.70 and 0.95 Å is actually a short linear extrapolation; this R range is essentially outside the range of the data, as indicated by ψ^2 for the $\nu_X=20$, $J=11$ state of ^{7}LiH, so that the highest order ϕ_n coefficients in Table II, which are influenced mostly by the potential in the outer limb toward long range, do not provide a reliable extrapolation at the smallest R values.

For the A state in Fig. 4, the unusually flat nature of the curve around R_e arises from the anomalous anharmonic form for the potential energy function, as discussed earlier. It should be mentioned that the apparent larger scatter very near R_e is an artifact that arises from rounding effects. As in Fig. 3, a small linear extrapolation was also needed at small R, in this case from 1.0 to 1.3 Å. As for the X state, the value of $R_{1/2}=6.00$ Å for the switching function corresponds to the largest R value sampled by the data, as indicated by ψ^2 for the $\nu_A=20$, $J=17$ state of ^{7}LiH. The present results are compared with results calculated from the composite potential obtained by Stwalley *et al.*[57] and from the *ab initio* potential of Partridge *et al.*[61] Although the results in the long-range region from these latter studies show definite scatter, the actual energy differences are quite small; for $R=10.6$ Å, for example, the two points have binding energies of 1.69 and 1.96 cm^{-1}. It is gratifying that the curve found in the present work extrapolates so well into the long-range region ($R>7.4$ Å). It is clear that this analytical function provides a realistic description of the A-state potential over the entire well.

The correction functions $U^{\mathrm{Li}}(R)$ and $U^{\mathrm{H}}(R)$ for the A

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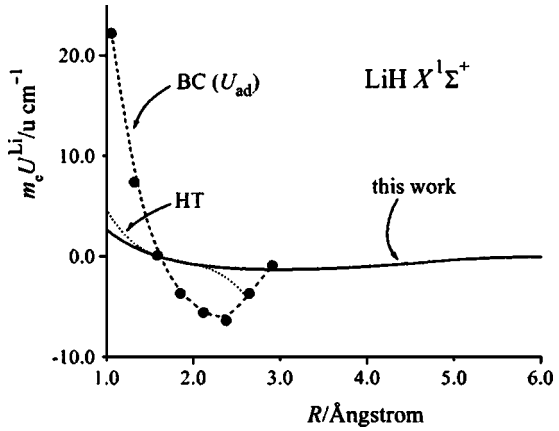


FIG. 5. $U^{Li}(R)$ for the $X\,^1\Sigma^+$ state of LiH. The solid curve is defined by the parameters listed in Table II. HT refers to the analytical function obtained by Hadinger and Tergiman (Ref. 14) from the expansion coefficients of Ref. 10, and BC refers to the *ab initio* calculation of Bishop and Cheung (Ref. 30) of the adiabatic correction function for the Li center.

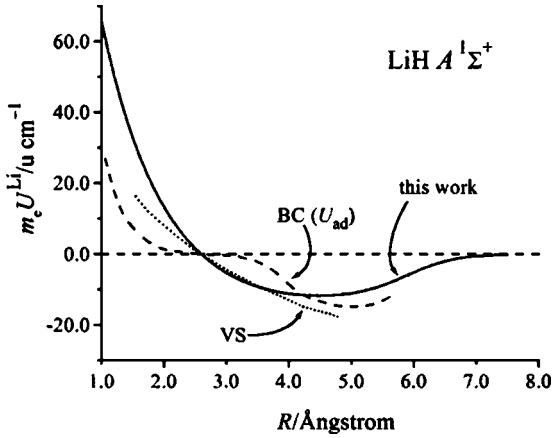


FIG. 7. $U^{Li}(R)$ for the $A\,^1\Sigma^+$ state of LiH. The solid curve is defined by the parameters listed in Table II. VS refers to the function determined experimentally by Vidal and Stwalley (Ref. 10); BC refers to the *ab initio* calculation of Bishop and Cheung (Ref. 30) of the adiabatic correction function for the Li center.

and X states of LiH, defined by the parameters in Table II, are shown in Figs. 5–8, together with experimental and theoretical results obtained previously. Figure 5 shows $U^{Li}(R)$ for the X state. This is the only function that was not determinable experimentally by Stwalley and co-workers.[10,13] However, the expansion coefficients of Ref. 10 were used by Hadinger and Tergiman[14] to obtain an analytical form for $U^{Li}(R)$, as indicated in Fig. 5. It is clear that there is good agreement for the slope at R_e between the function obtained in Ref. 14 and that of the present work. However, as indicated in Table II, our function is constrained so as to approach the asymptotic limit $U^{Li}(\infty)=0$, which corresponds to equal values of D_e for ^{6}LiH and ^{7}LiH, as well as for ^{6}LiD and ^{7}LiD. The slopes at R_e of the experimental curves are significantly different than that of the adiabatic correction function from the *ab initio* calculations of Bishop and Cheung.[30] The corresponding results for $U^H(R)$ in the X state are shown in Fig. 6. In this case, the points from the IPA analysis of Vidal and Stwalley,[10] and especially that of Chan *et al.*,[13] are in excellent agreement with the curve found in the present work, which is constrained to the limit $m_e U^H(\infty) = -11.6$ u cm^{-1}, corresponding to the 5.8 cm^{-1} larger dissociation energy of LiD(X) over LiH(X). Except at the largest R values, the analytical function of Ref. 14 agrees well with the results of Vidal and Stwalley.[10] The theoretical adiabatic correction function[30] is also shown and is quite similar to the experimental $U^H(R)$ correction functions, and more so for the similar comparison in Fig. 5 for the Li center.

Information available for the $U^{Li}(R)$ function of LiH(A) is shown in Fig. 7. As for the X state, the function determined in the present work is switched such that $U^{Li}(\infty)=0$. Near R_e, there is fair agreement with the function found by Vidal and Stwalley.[10] However, the experimental curves differ significantly from the theoretical adiabatic correction function calculated by Bishop and Cheung.[31] The opposite situation is found in Fig. 8 for $U^H(R)$ in the A state. Here the *ab initio*

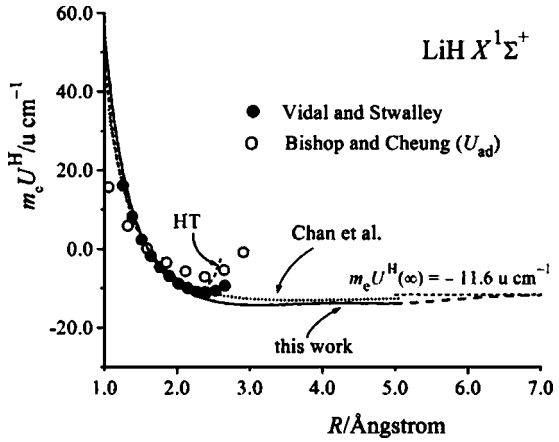


FIG. 6. $U^H(R)$ for the $X\,^1\Sigma^+$ state of LiH. The solid curve is defined by the parameters listed in Table II and is constrained to the limit $m_e U^H(\infty) = -11.6$ u cm^{-1}, as explained in the text. The results from the IPA analyses of Vidal and Stwalley (Ref. 10) and Chan *et al.* (Ref. 13), as well the analytical function of Hadinger and Tergiman (Ref. 14) (denoted as HT) and the adiabatic correction obtained theoretically by Bishop and Cheung (Ref. 30), are also shown.

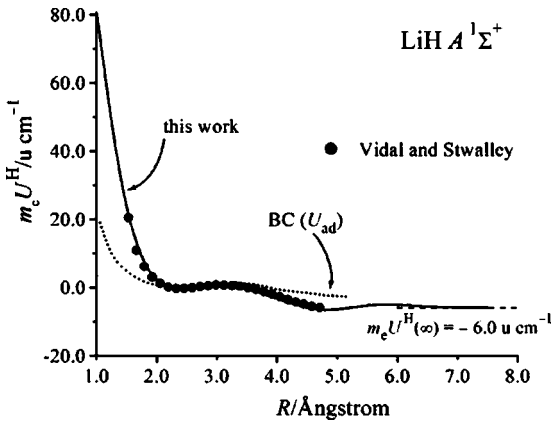


FIG. 8. $U^H(R)$ for the $A\,^1\Sigma^+$ state of LiH. The solid curve is defined by the parameters listed in Table II and is constrained to the limit $m_e U^H(\infty) = -6.0$ u cm^{-1}, as explained in the text. The results from the IPA analysis of Vidal and Stwalley (Ref. 10) and the adiabatic correction obtained theoretically by Bishop and Cheung (Ref. 31) are also shown.

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TABLE III. Vibrational terms and rotational constants for the $X\,^1\Sigma^+$ state of LiH. J_m denotes the maximum value of J for the data set of the present work; an asterisk is used for vibrational levels not sampled by the data set.

	^{6}LiH			^{7}LiH			^{6}LiD			^{7}LiD		
ν	J_m	G_ν	B_ν	J_m	G_ν	B_ν	J_m	G_ν	B_ν	J_m	G_ν	B_ν
0	32	705.157 5	7.559 673 22	30	697.953 1	7.406 038 53	37	534.355 3	4.342 034 92	45	524.757 7	4.187 497 15
1	26	2 078.523 5	7.340 589 42	30	2 057.661 6	7.193 607 19	30	1 581.859 5	4.246 787 94	41	1 553.821 9	4.097 293 72
2	26	3 406.110 3	7.125 434 0	33	3 372.513 4	6.984 948 90	28	2 602.964 4	4.152 855 2	37	2 557.420 2	4.008 313 71
3	24	4 688.860 1	6.913 936 5	35	4 643.423 2	6.779 807 49	30	3 598.093 8	4.060 172 4	32	3 535.954 7	3.920 498 5
4	25	5 927.63	6.705 8	42	5 871.229 3	6.577 893 90	32	4 567.647	3.968 673 1	20	4 489.804 2	3.833 787 8
5	23	7 123.20	6.500 6	39	7 056.681 2	6.378 859 78	26	5 511.995	3.878 281 8	17	5 419.323 1	3.748 113 5
6	19	8 276.23	6.298 0	34	8 200.425	6.182 284 5	24	6 431.481	3.788 911 0	14	6 324.838 4	3.663 396 6
7	15	9 387.24	6.097 4	29	9 302.99	5.987 66	23	7 326.41	3.700 46	12	7 206.65	3.579 55
8	12	10 456.62	5.898 1	25	10 364.75	5.794 37	*	8 197.07	3.612 8	24	8 065.01	3.496 46
9	*	11 484.56	5.699 2	32	11 385.92	5.601 63	*	9 043.67	3.525 8	24	8 900.14	3.414 01
10	12	12 471.03	5.499 9	27	12 366.48	5.408 45	*	9 866.39	3.439 4	24	9 712.24	3.332 06
11	*	13 415.70	5.298 5	20	13 306.14	5.213 54	*	10 665.37	3.353 2	24	10 501.42	3.250 45
12	12	14 317.84	5.093 3	11	14 204.25	5.015 22	*	11 440.66	3.267 2	8	11 267.75	3.168 98
13	*	15 176.25	4.882 0	*	15 059.71	4.811 3	*	12 192.23	3.180 9	24	12 011.23	3.087 42
14	12	15 989.08	4 661 2	11	15 870.82	4.598 8	*	12 919.97	3.094 2	*	12 731.77	3.005 48
15	*	16 753.69	4.426 8	11	16 635.15	4.373 9	*	13 623.65	3.006 6	24	13 429.17	2.922 80
16	12	17 466.38	4.172 9	11	17 349.28	4.131 4	*	14 302.89	2.917 6	12	14 103.13	2.838 97
17	*	18 122.05	3.891 6	11	18 008.52	3.864 0	*	14 957.14	2.826 6	24	14 753.17	2.753 45
18	*	18 713.73	3.571 4	*	18 606.48	3.561 6	*	15 585.66	2.732 8	24	15 378.65	2.665 61
19	12	19 231.91	3.195 5	11	19 134.44	3.209 4	*	16 187.46	2.635 4	24	15 978.72	2.574 64
20	12	19 663.85	2.740 7	11	19 580.70	2.786 2	*	16 761.24	2.533 1	12	16 552.24	2.479 56
21	11	19 994.15	2.185 1				*	17 305.33	2.424 4	8	17 097.79	2.379 14
22							*	17 817.64	2.307 5	24	17 613.54	2.271 81
23							*	18 295.47	2.179 9	*	18 097.16	2.155 53
24							*	18 735.42	2.038 3	24	18 545.76	2.027 65
25							*	19 133.21	1.878 3	24	18 955.65	1.884 63
26							*	19 483.44	1.694 3	8	19 322.22	1.721 83
27										24	19 639.79	1.533 56

result is remarkably similar around R_e to the experimental curve found herein and with the points from Ref. 10. As indicated in Table II and Fig. 8, the switching function leads to $m_e U^H(\infty) = -6.0\,\mathrm{u\,cm^{-1}}$, corresponding to the 3.0 $\mathrm{cm^{-1}}$ larger dissociation energy of LiD over LiH in the A state.

It should be emphasized that a direct equivalence between the experimental correction functions and the theoretical adiabatic correction functions of Figs. 5–8 should not be assumed. The experimental $U^k(R)$ functions could contain, in some unknown proportion, a contribution from nonadiabatic BO-breakdown effects. In this context, there is no obvious explanation for relatively close similarity in Figs. 6 and 8 between the experimental and adiabatic[30,31] $U^H(R)$ functions for the X and A states, but the quite different forms for the $U^{Li}(R)$ functions in Figs. 5 and 7. Further theoretical work would be of much interest.

One of the main advantages of the DPF method is that there is no involvement with intermediate sets of Dunham parameters, Y_{kl}. In traditional analyses of the spectra of diatomic hydrides, it is often necessary to consider Y_{kl} of such high order that correlations and truncation always lead to parameter estimates for which the physical significance is somewhat doubtful. The need for polynomial expansions with terms involving high order Y_{kl} is exemplified by the following considerations for LiH(X).

From the effective potential energy function for ^{7}LiH(X), as defined in Table II, together with the $q(R)$ function for this isotopomer, we have calculated the quantum mechanical molecular constants B_ν, D_ν, H_ν, L_ν, M_ν, N_ν, and O_ν for $\nu=0$–20. The calculations were performed using a version of Le Roy's program LEVEL (Ref. 62) that was modified to include the effect of $q(R)$. These constants were used to calculate vibration-rotational term values; for each vibrational level, the calculations extended up to the maximum J value sampled by the present data set and employed the standard form,

$$T(\nu,J) = G_\nu + B_\nu x - D_\nu x^2 + H_\nu x^3 + \cdots + O_\nu x^7, \qquad (17)$$

where $x = J(J+1)$. These terms, based on the centrifugal distortion constants (CDC) for the effective potential determined in the present work, were then compared with the quantum mechanical term values output from the DPF analysis. It was found that the term values calculated from the CDCs were reliable to within 0.000 1, 0.001, 0.01, and 0.1 $\mathrm{cm^{-1}}$ for J values up to ≈19, ≈23, ≈26, and ≈31, respectively. For $J=42$ in $\nu=4$, the error was about 4.6 $\mathrm{cm^{-1}}$. Similar results were found for other isotopomers. Since it is clear, then, that high order expressions are required in order to represent the ground state term values of LiH, in some cases with additional terms over those of Eq. (17), there is a clear preference for the DPF analysis, for which only 27 fitted parameters are required to represent accurately the rotational terms for all vibrational levels and for all four isotopomers. The parameters G_ν and B_ν are listed in Table III; the full set of calculated CDCs is available as supplementary data.[60]

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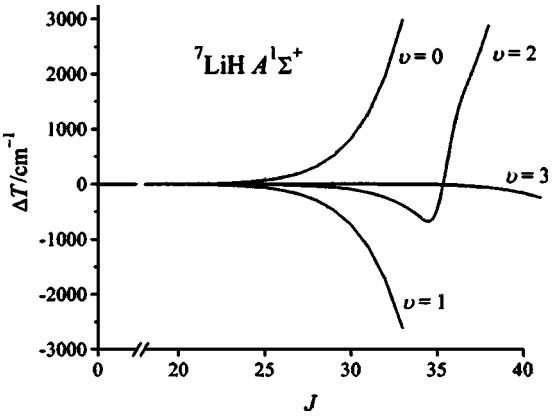


FIG. 9. The J dependence of the differences $\Delta T = T_{\mathrm{CDC}}(\nu,J) - T_{\mathrm{QM}}(\nu,J)$ for the $A\,^1\Sigma^+$ state of ^{7}LiH. $T_{\mathrm{CDC}}(\nu,J)$ are term values generated by a set of G_ν, B_ν, and centrifugal distortion constants D_ν–O_ν, calculated from the potential curve and $q(R)$ function of the DPF analysis. $T_{\mathrm{QM}}(\nu,J)$ are the quantum mechanical term values of the DPF analysis.

A similar consideration of the CDCs for the A state of LiH is particularly insightful. As mentioned earlier, in their analysis of the extensive A-X data set for ^{7}LiH, Orth and Stwalley[9] showed that a representation of the type given in Eq. (17), but with $H_\nu x^3$ as the highest order term, while leading to a generally satisfactory fit for most ν levels of the A state, was quite unsatisfactory for the lowest levels, $\nu=0$–2. It was necessary to truncate the data for these levels, such that only the lower-J data were considered. Orth and Stwalley[9] concluded that this anomalous behavior was attributable to the unusual and highly anharmonic shape of the A state potential. This is now confirmed herein by a CDC calculation for ^{7}LiH(A) similar to that for the X state. The discrepancies between $T_{\mathrm{CDC}}(\nu,J)$, the CDC-generated term values using Eq. (17), and $T_{\mathrm{QM}}(\nu,J)$, the quantum mechanical term values from the present DPF analysis, are displayed in Fig. 9 for $\nu=0$–3. It is clear that massive discrepancies set in at $J\approx 20$ for $\nu=0$ and 1. Similar discrepancies become evident for $\nu=2$ at $J>27$ and for $\nu=3$ at $J>35$. Given the results in Fig. 9, it is not surprising that data at high J could not be included in the fits of Ref. 9.

The results in Fig. 9 employ Eq. (17) with CDC terms up to $O_\nu x^7$. It is of interest to examine how the sets of CDCs perform as the order of the expansion of Eq. (17) is changed. Table IV lists the residuals $T_{\mathrm{CDC}}(\nu,J) - T_{\mathrm{QM}}(\nu,J)$ for a selected J value in each of the levels $\nu=0$–12. For $\nu=0$–2, the values of J are those found by Orth and Stwalley[9] above which truncation of the data sets was considered necessary. For $\nu=3$–12, the values of J correspond to the maximum J values sampled by the data of the present DPF analysis. Table IV shows that with the truncations at $J=22$ and 29 for $\nu=1$ and 2, *none* of the sets of CDCs gives a satisfactory result; truncation is needed at even lower J values. To some extent, the fitted parameters B_ν, D_ν, and H_ν of Ref. 9 would become effective in character to accommodate the discrepancies shown in the fourth column of Table IV. It is clear that the conclusion of Orth and Stwalley[9] regarding the inadequacy of a polynomial expansion in $J(J+1)$ for the lowest ν levels of LiH(A) is confirmed by the present work. Furthermore, Table IV indicates that the same problem for $\nu=3$ is actually much more severe; the $J=41$ level is sampled by the 3–4 band measured by Li and Stwalley.[46] These authors performed direct least-squares fits for individual bands, but did not mention encountering any difficulty that would require data truncation. Unfortunately, no mention is made in Ref. 46 of the standard deviations of the fits. For $J=41$, the H_ν term has a magnitude of $\approx 306\ \mathrm{cm}^{-1}$. In such a situation, the fitted parameters must necessarily take on a substantial effective character so as to accommodate the inadequacy of the model. A direct fit of the line positions in the 3-4 band performed in the present work confirms that a set of effective B_ν, D_ν, and H_ν parameters can perform surprisingly well. The standard deviation of the fit was $0.22\ \mathrm{cm}^{-1}$, only about three times larger than that achieved in the DPF analysis. However, the fitted value $H_3' = 8.75(14)\times 10^{-8}\ \mathrm{cm}^{-1}$, for

TABLE IV. Differences between vibration-rotational terms, $T_{\mathrm{CDC}}(\nu,J) - T_{\mathrm{QM}}(\nu,J)$, for the $A\,^1\Sigma^+$ state of ^{7}LiH. $T_{\mathrm{CDC}}(\nu,J)$ are term values generated from a set of calculated centrifugal distortion constants (CDC), see text; $T_{\mathrm{QM}}(\nu,J)$ are quantum mechanical term values obtained from the present DPF analysis. Increasingly high order sets of CDC were employed for the column entries from left to right, as indicated in the headings.

ν	J[a]	*BD*	*BDH*	*BDHL*	*BDHLM*	*BDHLMN*	*BDHLMNO*
0	15	−11.9	1.3	0.5	−0.3	0.02	0.04
1	22	−28.2	−5.5	−6.2	17.5	−7.5	−10.7
2	29	−76.2	−17.9	−36.4	67.6	13.4	−53.8
3	41	−420.6	−112.5	−247.5	952.6	−179.3	−242.0
4	34	−103.4	−21.2	−39.0	55.8	−2.0	−1.8
5	33	−71.6	−9.6	−20.3	19.7	−1.9	−1.3
6	21	−4.1	0.03	−0.2	0.04	0.003	0.009
7	21	−4.0	0.09	−0.15	0.03	−0.013	0.02
8	24	−8.6	0.28	−0.38	0.17	−0.19	0.18
9	22	−5.1	0.22	−0.17	0.12	−0.10	0.05
10	22	−5.1	0.32	−0.24	0.20	−0.13	0.04
11	17	−1.2	0.08	−0.04	0.016	−0.004	−0.000 1
12	19	−2.2	0.25	−0.14	0.07	−0.013	−0.005 4

[a]For $\nu=0$–2, the values of J are the maximum values employed in the fits of Orth and Stwalley (Ref. 9); for $\nu=3$–12, the values of J are the maximum values sampled by the data considered in the present work.

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example, is significantly larger than the CDC value calculated from the potential, $H_3' = 6.033 \times 10^{-8}\,\mathrm{cm}^{-1}$.

VI. CONCLUSIONS

In agreement with the conclusion of Orth and Stwalley,[9] it is confirmed that the standard Dunham polynomial expansion in $J(J+1)$ is an unsatisfactory model for the rotational states in the $A\,^1\Sigma^+$ state of LiH, particularly for the lowest vibrational levels with $v \leqslant 3$. With the *Direct Potential Fit* model employed in the present work, however, there is no anomalous problem in representing such states, and the entire manifold of vibration-rotational terms for the isotopomers ^{6}LiH, ^{7}LiH, ^{6}LiD and ^{7}LiD are reproduced simultaneously up to 84.9% of the well depth as the eigenvalues of standard $^1\Sigma$-state Hamiltonians containing an effective potential energy function and BO-breakdown functions. The simultaneous treatment of both the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states included all available line position data for the four isotopomers and provided estimates of the BO-breakdown functions $[q^{\mathrm{H,Li}}(R)$ and $U^{\mathrm{H,Li}}(R)]$. When the functions $U^{\mathrm{Li}}(R)$ and $U^{\mathrm{H}}(R)$, which contain contributions from adiabatic and nonadiabatic BO-breakdown effects in an unknown proportion, are compared with the *ab initio* adiabatic corrections $U^{\mathrm{Li}}_{\mathrm{ad}}(R)$ and $U^{\mathrm{H}}_{\mathrm{ad}}(R)$ for the X and A states, quite similar functions around R_e are found for the H center, but very different functions are found for the Li center. The acquisition of additional data for the higher vibrational levels of the X state for all four isotopomers would be very helpful; data of better precision for v levels already observed as well as data for higher levels near dissociation not yet studied are both needed.

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